

- 2 (a) The cube undergoes resonance when the driving frequency of the water wave equals the natural frequency of the oscillating system/cube. B1
There is maximum transfer of energy from the water wave to the cube/ The energy of the system/cube becomes a maximum and the system/cube oscillates with maximum amplitude. B1
- (b) $v = f\lambda$
 $f = 2.0 \text{ Hz}$ M1
Substitute $f = 2.0 \text{ Hz}$ into $f = \frac{1}{2\pi} \sqrt{\frac{g}{l}}$ A1
 $l = 0.0621 \text{ m}$
- (c) (i) Increase in wavelength results in decrease in driving frequency, thus driving frequency is not equal to natural frequency, Amplitude decreases. M1
A1
- (ii) Increase in mass results in an increase in l , and an decrease in natural frequency. Thus, driving frequency is not equal to natural frequency, Amplitude decreases. M1
A1
- (d) Drag force due to water on the cube causes damping. B1
Thus, maximum amplitude of oscillation occurs at a driving frequency 2 Hz, which is smaller than the natural frequency.
Since $f = \frac{1}{2\pi} \sqrt{\frac{g}{l}}$, a smaller value of f used in calculation results in a larger value of l calculated. B1
As such, the value determined in (b) is larger than actual measurement.

